

Restaurant Menu Nutrition Dataset Write-up

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Task 1: Observation and Data Collection Plan

Our group is interested in comparing restaurant menu items based on their nutritional content, price, and food characteristics. We chose this topic because menu items that seem similar can have very different macros like calories, protein, carbohydrates, fat, sugar, sodium, serving size, and price. We wanted to organize these characteristics into a dataset that would let us delve deeper into those differences. One observation represents one menu item from one restaurant.

We planned to collect items from seven restaurants: McDonald's, Taco Bell, Wendy's, Panera Bread, Chipotle, Chili's, and The Cheesecake Factory. These restaurants represent fast food, fast casual, and casual dining, giving us a good variation across restaurant types rather than putting our sole focus on one genre of restaurant. We also wanted to include a mix of food categories such as burgers, sandwiches, salads, breakfast items, bowls, tacos, burritos, and sweets.

Initial Domain Questions

1. What food item provides the best protein?
2. How does nutritional content differ between fast-food, fast-casual, and casual-dining restaurants?
3. What relationship exists between the price of a menu item and its nutritional content, especially protein and calories?
4. How do vegetarian and non-vegetarian menu items differ nutritionally?
5. How does serving size relate to calories and other nutritional values?

Data Collection Plan

We manually collected information from official restaurant websites and online nutrition menus instead of downloading an already prepared dataset. Group members were responsible for different restaurants and recorded their observations using the same set of attributes. For each item, we recorded restaurant, restaurant type, food category, menu item, dietary type, main ingredient, calories, protein, carbohydrates, fat, sugar, sodium, serving size, and price.

Our sampling focused on categorical coverage. We planned to collect roughly similar numbers of items from each restaurant and include different food categories so that one restaurant or food type would not dominate the dataset. Using three restaurant types also let us make deeper comparisons.

We knew that this approach won't let us get every item on every menu and that restaurants with detailed nutrition information available online would be easier to include. Prices could also vary by location, and different websites could report serving sizes or nutritional information differently. Because different group members were collecting the data, we planned a small sample before completing the dataset to make sure our definitions and recording methods were the same.

Initial Data Dictionary

Attribute	Type	Description	Example
Restaurant	Categorical	Name of the restaurant offering the menu item	McDonald's
Restaurant Type	Categorical	Classification of the restaurant based on its service style	Fast Food
Food Category	Categorical	General type of food the menu item belongs to	Burger
Menu Item	Categorical	Name of the specific menu item	Big Mac
Dietary Type	Categorical	Classification of the item as vegetarian or non-vegetarian	Non-Vegetarian
Main Ingredient	Categorical	Primary ingredient or protein source of the menu item	Beef
Calories (Cal)	Quantitative	Total calories in one serving of the menu item	550 Cal
Protein (g)	Quantitative	Amount of protein in one serving	25 g
Carbohydrates (g)	Quantitative	Amount of carbohydrates in one serving	46 g
Fat (g)	Quantitative	Amount of total fat in one serving	30 g
Sugar (g)	Quantitative	Amount of sugar in one serving	9 g
Sodium (mg)	Quantitative	Amount of sodium in one serving	1050 mg
Serving Size (g)	Quantitative	Weight of one serving of the menu item	220 g
Price (\$)	Quantitative	Listed price of the individual menu item	\$5.99

Example of source of observation from raw data (refer to ArtifactExampleRawData for more examples)



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	Serving Size	Calories (kcal)	Calories from Fat (kcal)	Fat (g)	Saturated Fat (g)	Trans Fat (g)	Total Cholesterol (mg)	Sodium (mg)	Total Carbohydrate (g)	Total Dietary Fiber (g)	Total Sugars (g)	Protein (g)	Calories from Alcohol (kcal)
Fuji Apple Chicken Salad without pecans - Whole	1 Salad	600	310	34	7	0	85	1770	47	4	31	26	N/A
Greek Salad - Half	1/2 Salad	310	270	30	6	0	15	700	9	2	4	4	N/A
Greek Salad - Whole	1 Salad	630	540	59	12	0	30	1400	17	5	7	8	N/A
Greek Salad with Chicken - Half	1/2 Salad	380	280	31	6	0	50	1300	11	3	5	15	N/A
Greek Salad with Chicken - Whole	1 Salad	760	560	62	13	0	100	2600	22	5	9	30	N/A
Green Goddess Cobb Salad with Chicken - Half	1/2 Salad	290	150	17	3.5	0	140	990	15	4	8	20	N/A
Green Goddess Cobb Salad with Chicken - Whole	1 Salad	580	310	34	7	0	280	1980	30	7	15	40	N/A
Mediterranean Chicken Greens with Grains Salad - Half	1/2 Salad	450	290	32	5	0	40	1180	24	4	4	16	N/A
Mediterranean Chicken Greens with Grains Salad - Whole	1 Salad	900	580	64	11	0	85	2330	47	7	8	33	N/A
Mediterranean Greens with Grains Salad - Half	1/2 Salad	380	280	31	5	0	10	550	21	3	3	5	N/A
Mediterranean Greens with Grains Salad - Whole	1 Salad	760	560	62	10	0	15	1100	42	7	6	11	N/A
Ranch Parm BLT Salad- Half	1/2 Salad	380	270	30	9	0	45	950	14	2	6	12	N/A
Ranch Parm BLT Salad- Whole	1 Salad	750	550	61	17	0.5	85	1900	29	4	11	24	N/A
Mediterranean Chicken Greens with Grains Salad Stuffer	1 Stuffer	840	470	52	8	0	40	1810	66	5	6	24	N/A
Mediterranean Greens with Grains Salad Stuffer	1 Stuffer	770	460	51	8	0	10	1200	63	5	5	13	N/A
Ranch Parm BLT Salad Stuffer	1 Stuffer	770	460	51	12	0.5	60	1740	59	3	9	21	N/A
Santa Fe Chicken Salad Stuffer	1 Stuffer	740	390	43	8	0	75	1930	65	5	13	24	N/A

Task 2: Pilot and Data Collection

We started with a pilot of about 10 menu items to test our collection process. Most numerical attributes were fairly straightforward, but restaurant websites sometimes showed nutrition and serving size information differently in raw form. Food category, dietary type, and main ingredient were more difficult because some items could reasonably fit into multiple categories or contain several important ingredients. Since group members were collecting different restaurants, the pilot also showed that the same type of item could be classified differently without having clear rules.

Based on the pilot, we standardized our category names, units, and definitions and made sure that one row always represented one individual menu item. We did not find a major attribute that needed to be added or removed, but we made some attributes clearer that were a little vague in the data dictionary so they could be recorded more consistently. The pilot also showed that the dataset could support more than simple high vs low comparisons, which influenced the questions we considered later.

After these changes, we increased the dataset to 98 menu items from 7 restaurants. We kept the number of observations from each restaurant pretty similar and included multiple food categories across fast food, fast casual, and casual dining. This was more useful than simply collecting as many items as possible from one restaurant because it gave us more variety across restaurant type, food category, nutrition, serving size, and price.

The final observations were combined into one machine readable spreadsheet. The dataset represents a snapshot of the menus during our collection period rather than every item or

every possible restaurant location. Our final data dictionary kept the same 14 attributes but used the standardized definitions and units made after the pilot.

Task 3: Data Description and Domain Questions

Our final dataset contains 98 menu items from 7 restaurants and 14 attributes: restaurant, restaurant type, food category, menu item, dietary type, main ingredient, calories, protein, carbohydrates, fat, sugar, sodium, serving size, and price. The dataset mainly has categorical coverage, with variation across three restaurant types, multiple food categories, dietary types, ingredients, and a wide range of nutritional values, serving sizes, and prices. Because the restaurants were selected from chains with nutrition information available online rather than through random sampling, the dataset isn't a representation of all restaurants. Some food categories also have more observations than others. Prices may vary by location, and serving sizes were not always reported in exactly the same way. Mixed dishes were especially difficult to classify because they could contain several major ingredients or fit more than one food category.

Turning restaurant menus into structured data also required us to simplify the real world items. We decided that each row would represent one menu item and that each item would receive one food category, dietary type, and main ingredient classification. This made comparison possible, but some information was lost, including every individual ingredient, customization options, taste, menu changes over time, and exact location differences.

Our questions changed as the dataset developed. At first, we were mainly thinking about simple questions such as which restaurant has the most protein or calories. After looking at the full dataset, we realized that those rankings could hide important context. For example, serving size can affect total calories, and comparing price and protein can reveal a different pattern than looking at either attribute alone. We also noticed that some food categories contained unexpected nutritional values. This led us to replace several of our original questions with questions that focus more on distributions, relationships, and unexpected patterns.

Final Domain Questions

1. How do calories, protein, and sodium differ across restaurant types?

This question uses Restaurant Type, Calories, Protein, and Sodium to examine whether broader nutritional patterns differ between fast food, fast casual, and casual dining rather than simply ranking individual restaurants.

2. How does serving size relate to calories across menu items, and does this relationship differ across restaurant types?

This question uses Serving Size, Calories, and Restaurant Type. It gives more context to calorie

differences by considering whether larger portions are associated with higher total calories and whether that relationship is similar across restaurant types.

3. How does price relate to protein, and does this relationship differ across restaurant types?

This question connects Price, Protein, and Restaurant Type. Instead of simply finding the cheapest or highest-protein item, it explores whether higher prices are associated with more protein and whether that pattern changes across types of restaurants.

4. How is sugar distributed across food categories, and which non-sweet items have unusually high sugar?

This question uses Sugar, Food Category, Restaurant, and Menu Item. It allows us to compare sugar distributions across categories while also looking for unexpected items that do not belong to the sweet or dessert category but still contain relatively high amounts of sugar.

These revised questions make better use of the variation we actually found during collection and give us different kinds of visualization problems rather than four versions of the same comparison.

Task 4: Task Abstractions

For the first question, the main goal is to compare and summarize different numerical values between different groups. We chose it this way because the goal is not to identify one highest value, but to understand the differences between the groups.

For the second question, the goal is to find and compare the relationship between two numerical values. We wanted to see if things like calories and serving size are connected and if that relationship changes between groups.

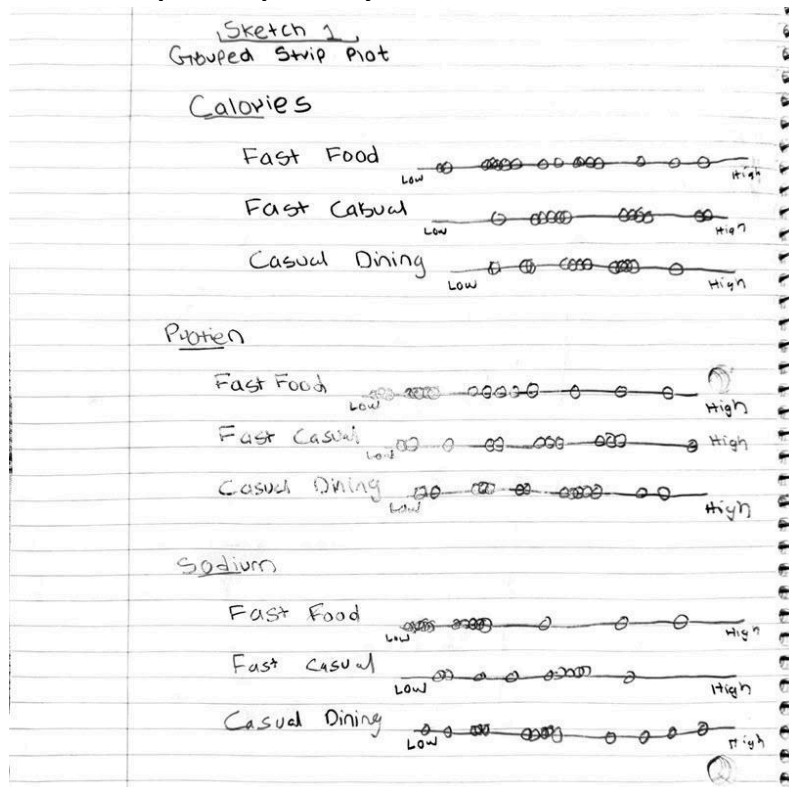
For the third question, we are also looking for a relationship between two values, like the price and protein. Instead of only finding which items would have the most protein or lowest price, we want to see if there is a pattern between price and protein and if that pattern would be different across groups.

For the fourth question, the objective is to compare groups and find anything that stands out. We want to see the normal range of values in each group and find items that are unusually high or low compared to the rest of the items.

Overall, task abstraction helped us focus on what we are actually trying to learn from the data before deciding what kind of graph to use. It helped us think about comparing groups, finding relationships between values, and noticing unusual values or outliers.

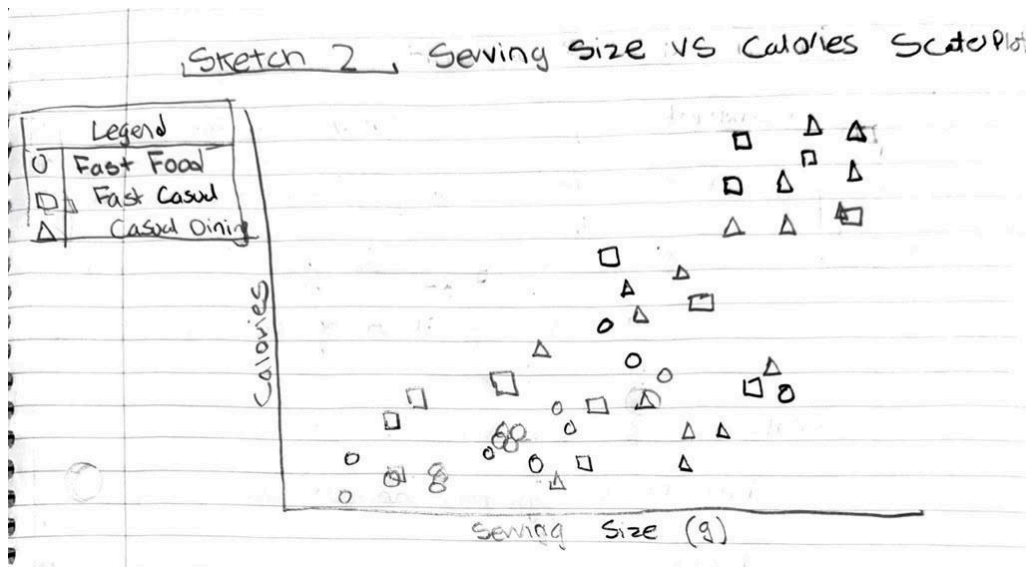
Task 5: Visualization sketches

Sketch 1 – Grouped Strip Plot by Omais



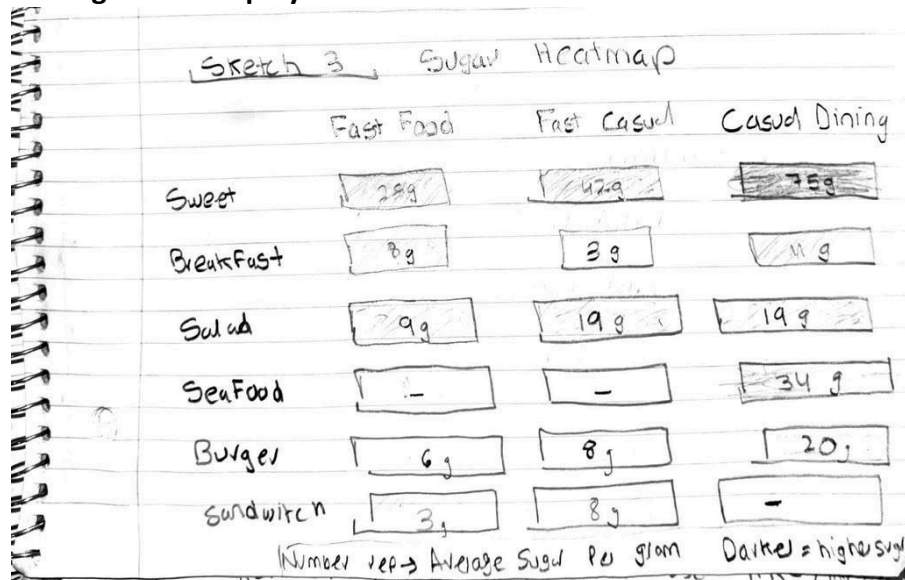
The goal of this sketch was to compare how calories, protein, and sodium vary across restaurant types instead of only comparing averages (Q1). It helps with showing the differences between fast food, fast casual, and casual dining restaurants. The main attributes are restaurant type, calories, protein, and sodium. Each menu item is shown as a point, with horizontal position representing the nutrition value and shape representing restaurant type. This makes it easier to see the overall range of values and notice any items that stand out, but the different nutrition measures use different scales, which makes having a direct comparison harder. This sketch differs from the others because it focuses on distributions of individual items

Sketch 2 – Serving Size vs. Calories Scatterplot by Omais



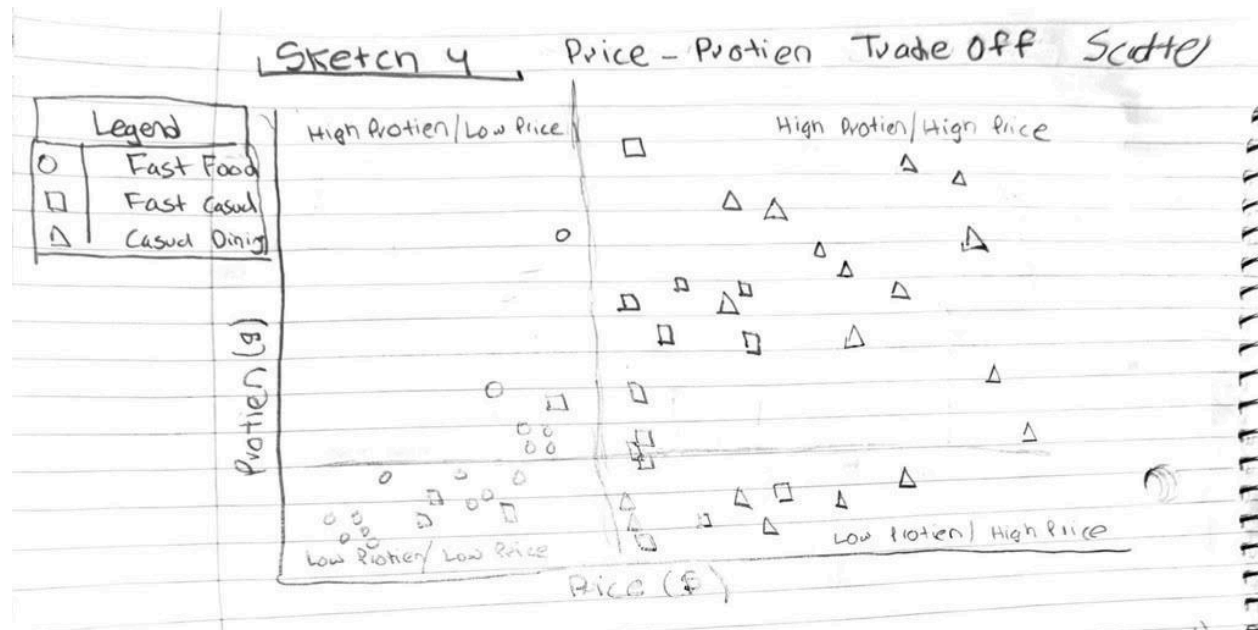
The goal of this sketch was to see if larger serving sizes are connected to higher calories (Q2). Serving size is shown on the x-axis and calories are shown on the y-axis, while different shapes represent the restaurant types. Each point represents a menu item, making it easy to see the overall pattern and any items that stand out. One drawback is that the graph does not show whether an item has more calories because of its larger serving size or because the food itself is higher in calories. This sketch is different from Sketch 1 because it focuses on the relationship between two values instead of how the values are spread out.

Sketch 3 – Sugar Heatmap by Omais



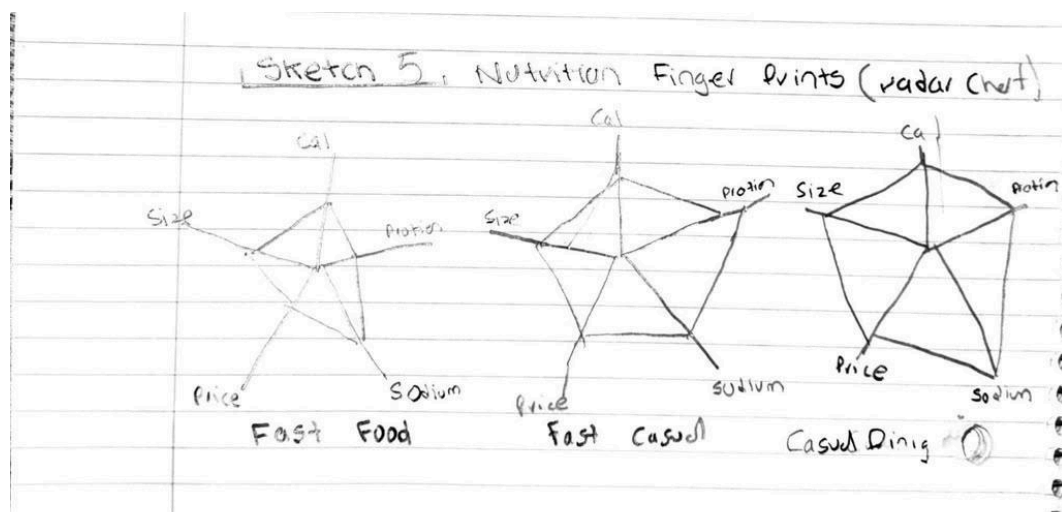
The goal of this sketch was to compare sugar levels across different food categories and restaurant types. It helps show which categories have higher or lower amounts of sugar and which ones stand out. The food categories and restaurant types are organized in rows and columns, while darker cells show higher average sugar levels. This makes it easy to quickly see overall patterns. One drawback of this is that using averages can hide individual menu items that have pretty high amounts of sugar. This sketch is different from the others because it uses a heatmap to summarize the data instead of showing each menu item separately.

Sketch 4 – Price-Protein Tradeoff Map by Muhammad



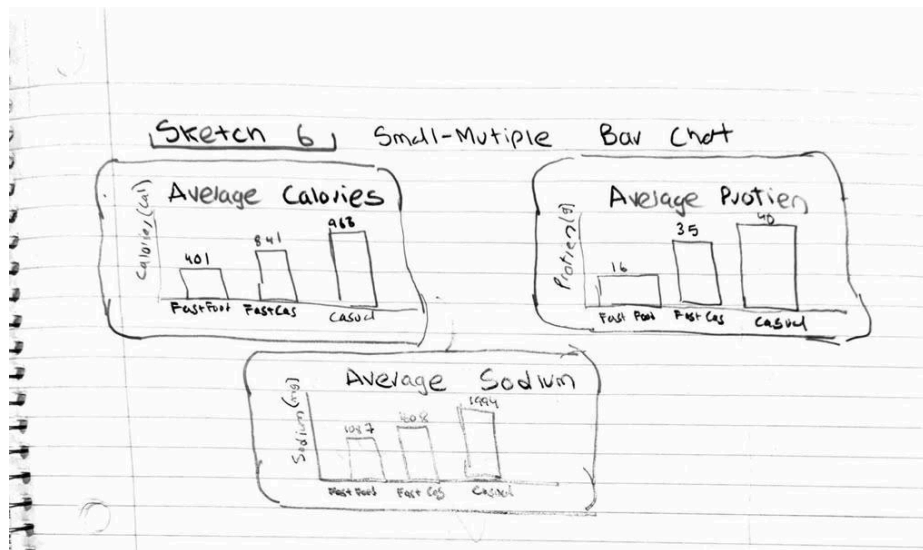
The goal of this sketch was to see if the more expensive an item is if it will also have a higher protein count, and to compare which items give you more protein for the price. Price is shown on the x-axis and protein on the y-axis, while different shapes represent restaurant types. The graph is split into sections, such as high-protein/low-price and low-protein/high-price, which makes it easier to see which items offer better value. One limitation is that deciding what counts as a high or low price or protein amount can be subjective. This sketch is different from a regular scatterplot because it focuses more on value and tradeoffs between price and protein.

Sketch 5 – Nutrition Fingerprints by Muhammad



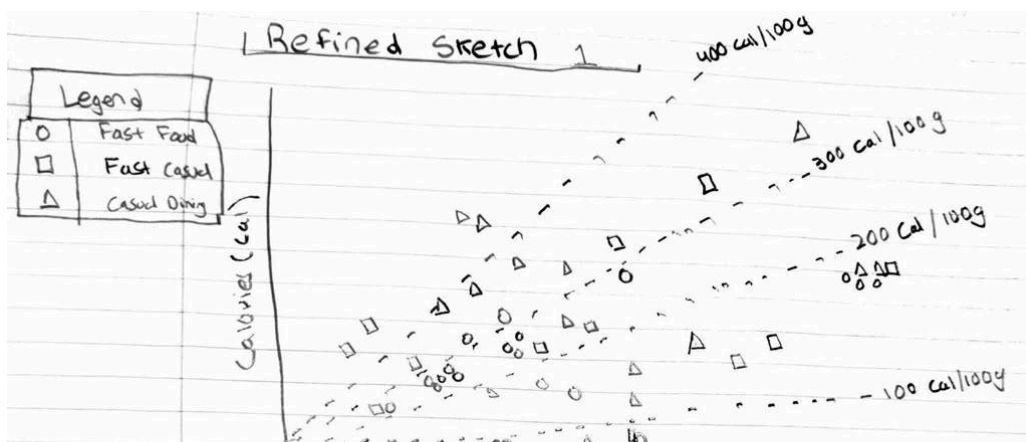
The purpose of this sketch was to compare several attributes at the same time and create an overall nutritional profile for each restaurant type. It uses calories, protein, sodium, price, and serving size. Each attribute is shown on a separate axis, and connected lines form a radar-shaped profile. Distance from the center represents the relative amount of each attribute. This makes the overall profiles visually easy to recognize, but exact values and small differences are difficult to compare. It differs from the other sketches because it represents many attributes in one multidimensional view.

Sketch 6 – Small-Multiple Bar Charts by Muhammad



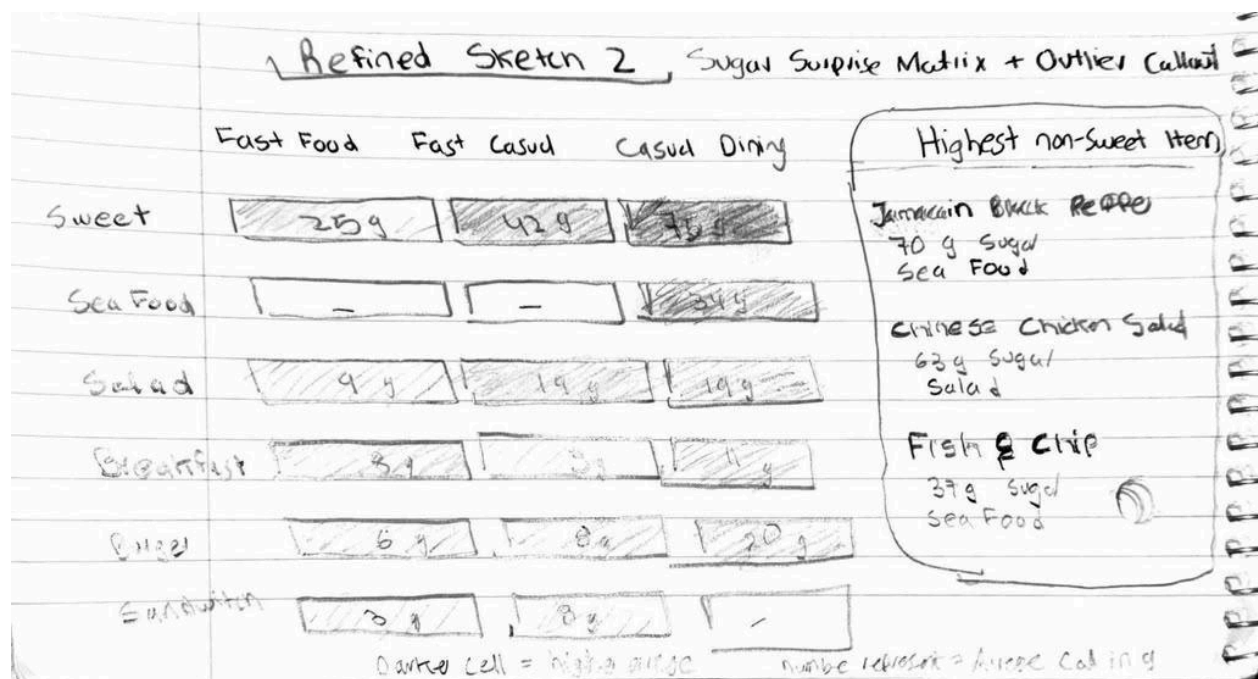
The goal of this sketch was to give a simple comparison between the different restaurant types. Separate bar charts show the average calories, protein, and sodium for fast food, fast casual, and casual dining restaurants. The length of each bar represents the average amount, making it easy to quickly compare the restaurant types. One limitation is that averages do not show the differences between individual menu items or any unusual items that stand out. This sketch is different from the others because it gives a simple overall summary instead of showing more detailed patterns in the data.

Refined Sketch 1 – Calorie-Density Scatterplot



This version improves on Sketch 2. The original graph compared serving size and calories, but it was hard to tell why some items had more calories. The new version adds calories per 100g lines, which helps show whether an item is high in calories because it has a larger serving size or because the food itself is more calorie dense. Each point still represents a menu item, and different shapes represent the restaurant types. This makes the graph more useful while still keeping the individual menu items visible.

Refined Sketch 2 – Sugar Heatmap with Outlier Callouts



This version improves on Sketch 3 by keeping the heatmap but also showing individual items that are unusually high in sugar. The heatmap still shows the average sugar levels for each food category and restaurant type, with darker areas representing higher sugar. Symbols or labels are added to point out non sweet items that have surprisingly high sugar. The original heatmap made it easy to see overall patterns, but using averages could hide items that stood out. This change makes it easier to see both the overall sugar patterns and the specific menu items that are unusual.

Task 6: Summarizing

We explored several different visualization ideas, including strip plots, scatterplots, heatmaps, radar charts, tradeoff maps, and bar charts. Some designs, like bar charts, were simple and easy to understand, but they did not show much detail about individual items. Other designs, like scatterplots and strip plots, showed more information but were a little more complicated to read. The heatmap worked well for comparing many categories, while the radar chart was more unique but harder to read accurately. Each visualization idea had its pros and cons.

Creating different sketches also changed how we looked at our questions. At first, we mostly thought about comparing averages, but the different designs helped us look at overall patterns, relationships between values, and items that stood out. This gave us different ways to answer our Task 3 questions instead of using the same type of graph for everything.

Each design was useful in a different way. Some were better for showing how values were spread out, while others were better for showing relationships or comparing categories. Since we collected the data ourselves, we also noticed that some categories had more items than others and some serving sizes were reported differently. If we collected the data again, we would try to have a more equal number of items in each food category.

Overall, exploring different designs helped us see the strengths and weaknesses of each one. Some were easier to understand but showed less detail, while others showed more information but were harder to read. This process helped us decide which ideas worked best and how we could improve them.

Task 7: Collaboration

We worked together through meeting in person, getting on phone calls, and text messages. We met on campus to talk about our dataset, decide what information we wanted to collect, and make sure we were recording everything the same way. Whenever we ran into a problem or were unsure about something, we talked about it together before making a decision.

We split the data collection pretty evenly, with each of us collecting about half of the restaurant data. We made sure to use the same categories, units, and labels so our data would stay consistent. We also shared our visualization ideas through text and compared our sketches with each other.

For the visualizations, we talked about different ideas together but created our own sketches individually. This helped us come up with a wider variety of designs. After finishing the sketches, we looked at the strengths and weaknesses of each one and decided together which ones we wanted to improve.

Overall, our collaboration worked well because we both contributed about the same amount of work. The main challenge was making sure the data we collected separately was consistent, but communicating with each other helped us fix those differences.